

Laporan Modul 4 — Networking Deep Dive: IP Addressing, Routing, Packet Analysis

Workshop Administrasi Jaringan · PENS — Teknik Informatika dan Komputer

- Kelompok: X = 2 · srv01 difungsikan sebagai **router internal**
- OS: Debian 13.4 (trixie) · Tanggal: 2026-06-12
- LAN-A: 10.10.2.0/24 (vboxnet0) · LAN-B: 10.20.2.0/24 (vboxnet1)

1. Ringkasan

srv01 dijadikan router antara dua subnet. cli01 dipindah ke LAN-B, IP forwarding diaktifkan di srv01, dan adm01 (LAN-A) diberi static route ke LAN-B. Komunikasi lintas subnet diuji dan dianalisis dengan tcpdump (ICMP & TCP handshake).

2. Topologi & IP Plan

Host	Interface	IP	Subnet	Gateway
srv01	enp0s8	10.10.2.11/24	LAN-A	- (router)
srv01	enp0s9	10.20.2.1/24	LAN-B	- (router)
adm01	enp0s8	10.10.2.10/24	LAN-A	route ke B
cli01	enp0s8	10.20.2.2/24	LAN-B	10.20.2.1

Diagram & logic lengkap: docs/routing_design_v1.md .

3. Routing & IP Forwarding (4)

- srv01 NIC3 ditambahkan (host-only vboxnet1), enp0s9 = 10.20.2.1/24.
- IP forwarding permanen: /etc/sysctl.d/99-router.conf → net.ipv4.ip_forward=1 (terverifikasi /proc/sys/net/ipv4/ip_forward = 1).
- cli01 pindah ke LAN-B (10.20.2.2) dengan gateway srv01.

Evidence: evidence/ip_forward_check.txt , evidence/route_table_srv01.txt .

4. Static Route di adm01 (5)

```
sudo ip route add 10.20.2.0/24 via 10.10.2.11
```

Dibuat persisten via `post-up` di `/etc/network/interfaces.d/lan0`. Uji: `ping 10.20.2.2` dari `adm01` → **0% loss, TTL=63** (lewat 1 hop router `srv01`).

5. Packet Analysis dengan tcpdump (6 & 7)

ICMP (`tcpdump -i enp0s8 -n icmp`): terlihat echo request `10.10.2.10` → `10.20.2.2` dan echo reply sebaliknya, lengkap dengan source/destination IP.

TCP handshake (`tcpdump -i any -n tcp port 22 and host 10.20.2.2`):

```
Flags [S]    = SYN
Flags [S.]   = SYN-ACK
Flags [.]    = ACK
```

Capture juga menunjukkan `srv01` me-route: paket masuk `enp0s8`, keluar `enp0s9`.

Evidence: `evidence/tcpdump_icmp.txt`, `evidence/tcpdump_ssh.txt`. Analisis (ARP, ICMP vs TCP, SYN, TTL): `docs/packet_analysis_notes.md`.

6. Troubleshooting Terstruktur (9)

Urutan diagnosa yang dipakai: cek IP (`ip a`) → route (`ip r`) → gateway → forwarding (`/proc/.../ip_forward`) → firewall → capture tcpdump. Tidak menebak firewall lebih dulu.

7. Definition of Done (Checklist)

Item	Status
<code>adm01</code> ↔ <code>cli01</code> beda subnet bisa ping	OK (TTL=63)
tcpdump menunjukkan ICMP	OK
tcpdump menunjukkan TCP handshake	OK (SYN/SYN-ACK/ACK)
Routing terdokumentasi jelas	OK
Alur paket source→destination dipahami	OK

8. Bukti Screenshot

[1] srv01 sebagai router (2 subnet + ip_forward=1 + routing table)

```
-- [1] srv01 sebagai ROUTER ==
enp0s8      UP      10.10.2.11/24 fe80::a00:27ff:fe5c:fab9/64
enp0s9      UP      10.20.2.1/24  fe80::a00:27ff:fe93:a195/64
ip_forward = 1
-- routing table --
10.10.2.0/24 dev enp0s8 proto kernel scope link src 10.10.2.11
10.20.2.0/24 dev enp0s9 proto kernel scope link src 10.20.2.1
fishlr@srv01:~$
```

[2] adm01 → cli01 lintas subnet (TTL=63 = via router)

```
== [2] adm01 (LAN-A) -> cli01 (LAN-B) lintas subnet ==
-- route ke LAN-B --
10.20.2.0/24 via 10.10.2.11 dev enp0s8
-- ping (TTL=63 = lewat router srv01) --
PING 10.20.2.2 (10.20.2.2) 56(84) bytes of data.
64 bytes from 10.20.2.2: icmp_seq=1 ttl=63 time=3.34 ms
64 bytes from 10.20.2.2: icmp_seq=2 ttl=63 time=1.61 ms
64 bytes from 10.20.2.2: icmp_seq=3 ttl=63 time=1.95 ms

--- 10.20.2.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2002ms
rtt min/avg/max/mdev = 1.608/2.298/3.341/0.749 ms
fishir@adm01:~$ _
```

[3] tcpdump ICMP (echo request/reply)

```
== [3] tcpdump ICMP di srv01 ==
[sudo] password for fishir:
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on enp0s8, link-type EN10MB (Ethernet), snapshot length 262144 bytes
01:32:14.415100 IP 10.10.2.10 > 10.20.2.2: ICMP echo request, id 13, seq 1, length 64
01:32:14.416110 IP 10.20.2.2 > 10.10.2.10: ICMP echo reply, id 13, seq 1, length 64
01:32:15.417032 IP 10.10.2.10 > 10.20.2.2: ICMP echo request, id 13, seq 2, length 64
01:32:15.418162 IP 10.20.2.2 > 10.10.2.10: ICMP echo reply, id 13, seq 2, length 64
01:32:16.417752 IP 10.10.2.10 > 10.20.2.2: ICMP echo request, id 13, seq 3, length 64
01:32:16.418848 IP 10.20.2.2 > 10.10.2.10: ICMP echo reply, id 13, seq 3, length 64
6 packets captured
6 packets received by filter
0 packets dropped by kernel
fishir@srv01:~$
```

[4] tcpdump TCP handshake (SYN / SYN-ACK / ACK)

```
== [4] tcpdump TCP handshake (SYN/SYN-ACK/ACK) ==
tcpdump: WARNING: any: That device doesn't support promiscuous mode
(Promiscuous mode not supported on the "any" device)
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on any, link-type LINUX_SLL2 (Linux cooked v2), snapshot length 262144 bytes
01:32:45.661474 enp0s8 In IP 10.10.2.10.33074 > 10.20.2.2.22: Flags [S], seq 599909156, win 64240, options [mss 1460,sackOK,TS val 2089825977 ecr 0,nop,wscale 7], length 0
01:32:45.661509 enp0s9 Out IP 10.10.2.10.33074 > 10.20.2.2.22: Flags [S], seq 599909156, win 64240, options [mss 1460,sackOK,TS val 2089825977 ecr 0,nop,wscale 7], length 0
01:32:45.662499 enp0s9 In IP 10.20.2.2.22 > 10.10.2.10.33074: Flags [S.], seq 4023507424, ack 599909157, win 65160, options [mss 1460,sackOK,TS val 2500715411 ecr 2089825977,nop,wscale 7], length 0
01:32:45.662524 enp0s8 Out IP 10.20.2.2.22 > 10.10.2.10.33074: Flags [S.], seq 4023507424, ack 599909157, win 65160, options [mss 1460,sackOK,TS val 2500715411 ecr 2089825977,nop,wscale 7], length 0
01:32:45.663433 enp0s8 In IP 10.10.2.10.33074 > 10.20.2.2.22: Flags [.], ack 1, win 502, options [nop,nop,TS val 2089825979 ecr 2500715411], length 0
01:32:45.663457 enp0s9 Out IP 10.10.2.10.33074 > 10.20.2.2.22: Flags [.], ack 1, win 502, options [nop,nop,TS val 2089825979 ecr 2500715411], length 0
6 packets captured
18 packets received by filter
0 packets dropped by kernel
fishir@srv01:~$
```

9. Penyimpangan dari Modul

- OS Debian 13 (trixie), bukan Debian 12.
- Konfigurasi jaringan via **ifupdown** (bukan nmcli) sesuai instalasi aktual; hasil akhir (IP statik, gateway, route) sama.
- User admin `fishir` menggantikan `student`.